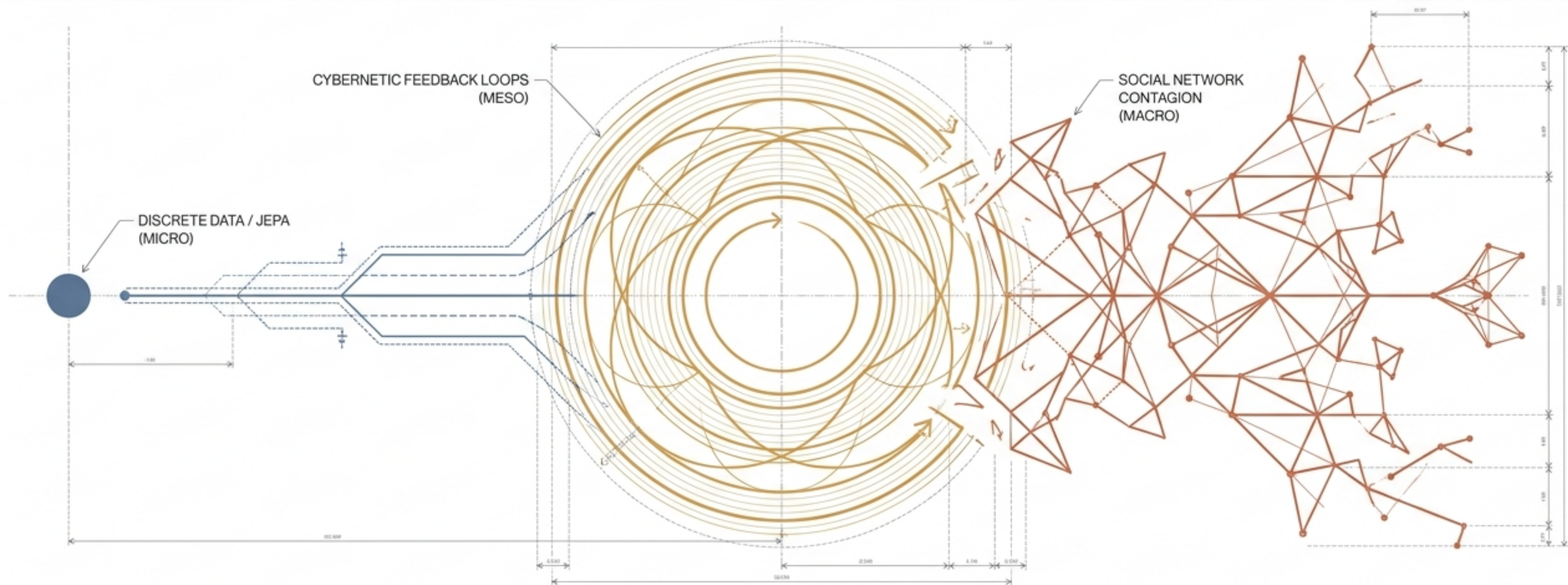


THE TOPOGRAPHY OF SIGNALS

FROM AI WORLD MODELS TO SYSTEMIC SOCIAL CONTAGION

A SYNTHESIS OF JOINT-EMBEDDING PREDICTIVE ARCHITECTURES,
SECOND-ORDER CYBERNETICS, AND SOCIAL SYSTEMS THEORY.

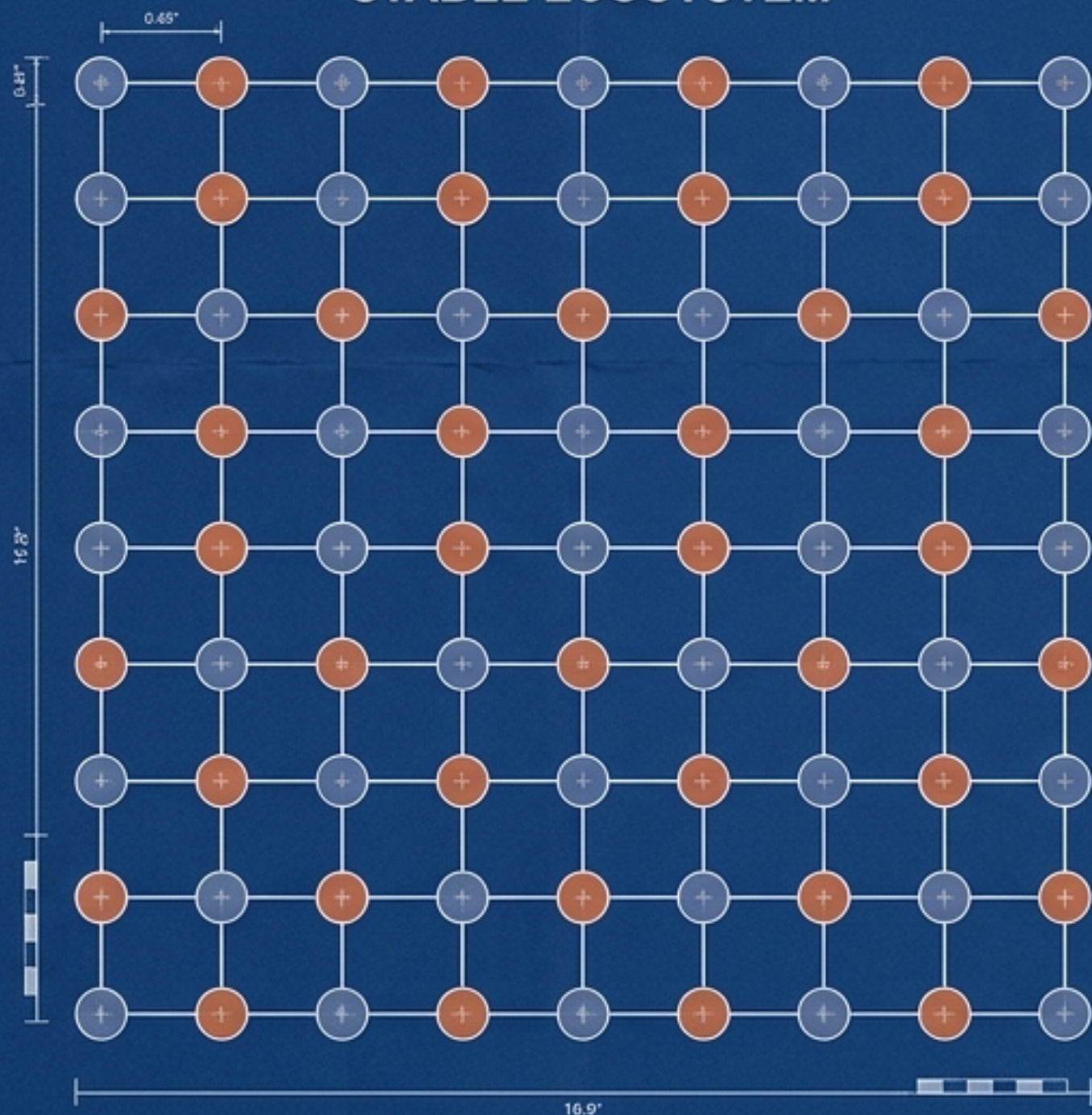


THE RISK OF EPISTEMIC COLLAPSE

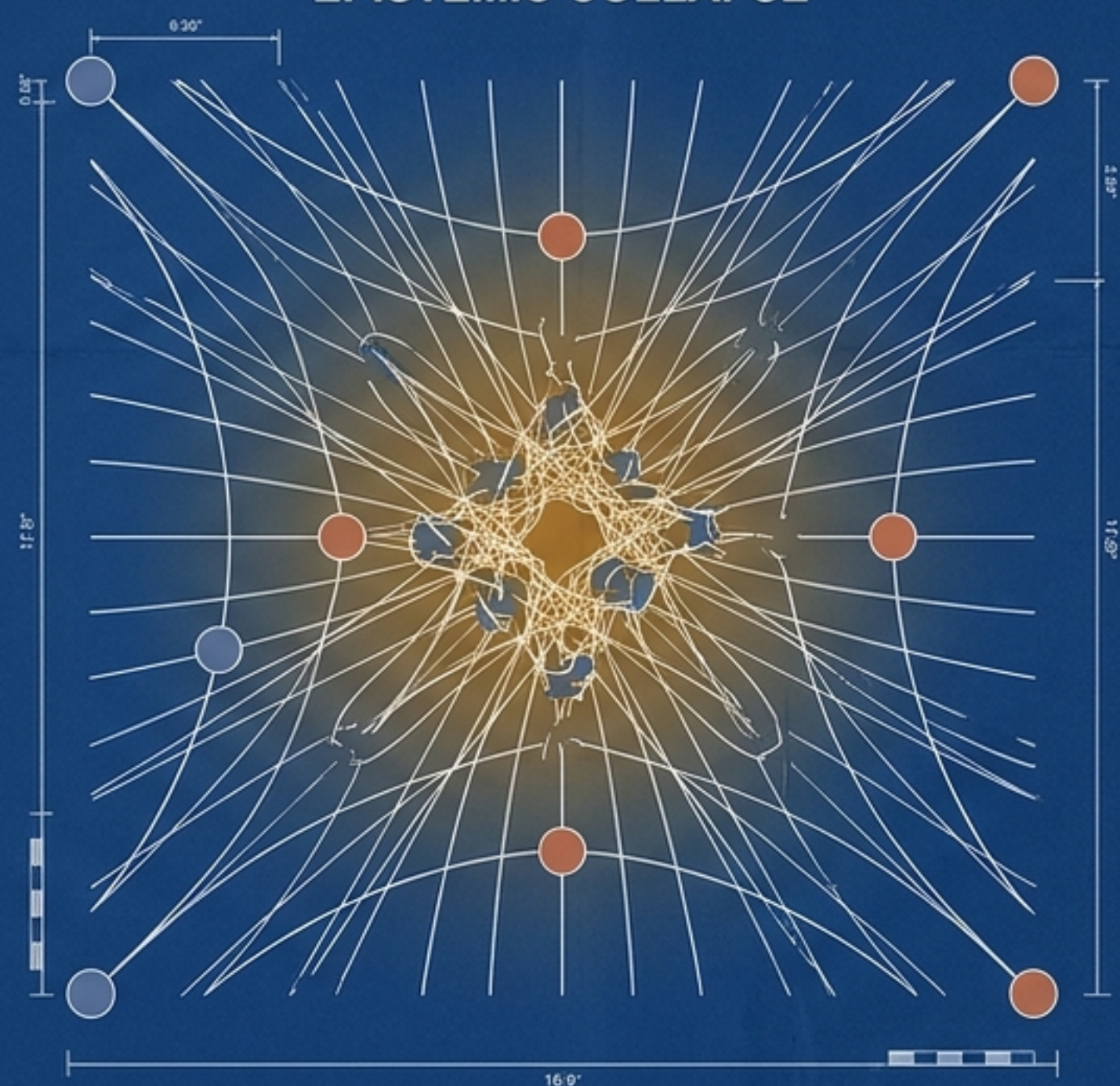
"When society becomes tightly coupled to algorithmic information production, small informational distortions endogenously destabilize the entire ecosystem."

Key Mechanism: Real-world social platforms are multiplex structures. When AI models train on the aggregate output of these platforms, local network biases are absorbed, amplified, and re-injected as objective reality.

STABLE ECOSYSTEM



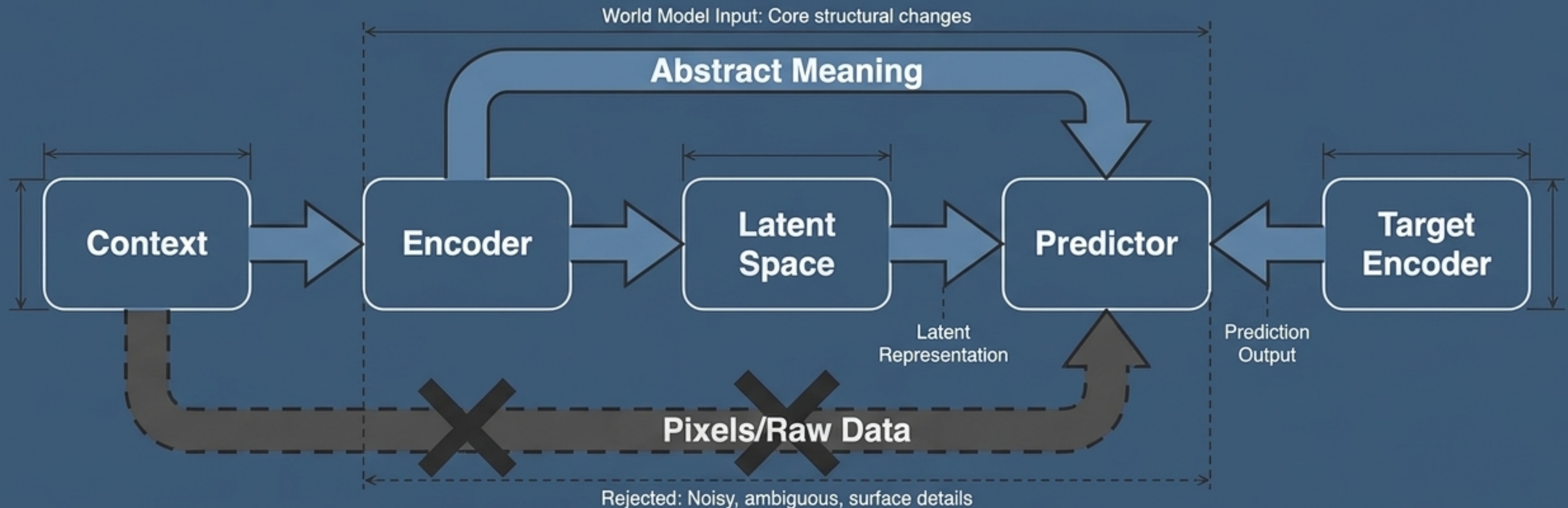
EPISTEMIC COLLAPSE



Level 1: The Architecture of Representation

The Joint-Embedding Predictive Architecture (JEPA)

Traditional AI generates the future pixel-by-pixel. JEPA predicts the future entirely in a latent representation space. By predicting abstract states rather than surface details, the system builds a "World Model"—an internal topography of how the world changes, discarding noisy or ambiguous data.



Generative vs. Predictive Topologies

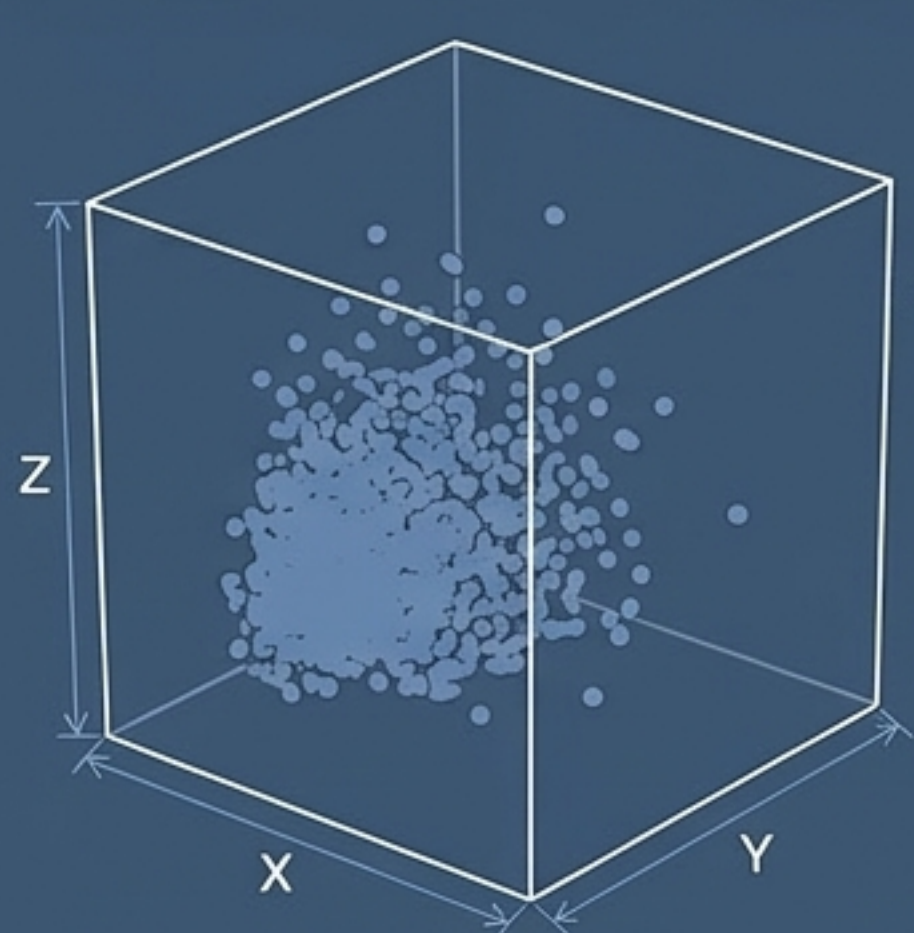
	Generative Models (LLMs / Pixels)	Predictive Models (JEPA)
Goal	Reconstruct Raw Data / Pixels	Predict Abstract Meaning
Space	Data Space	Latent Representation Space
Learning Mechanism	Comparison / Push-Pull	Time-travel / Future State Prediction
Primary Vulnerability	Hallucination & Blurry Prediction	Representation Collapse (mapping to a constant)

The Anatomy of a Healthy Distinction

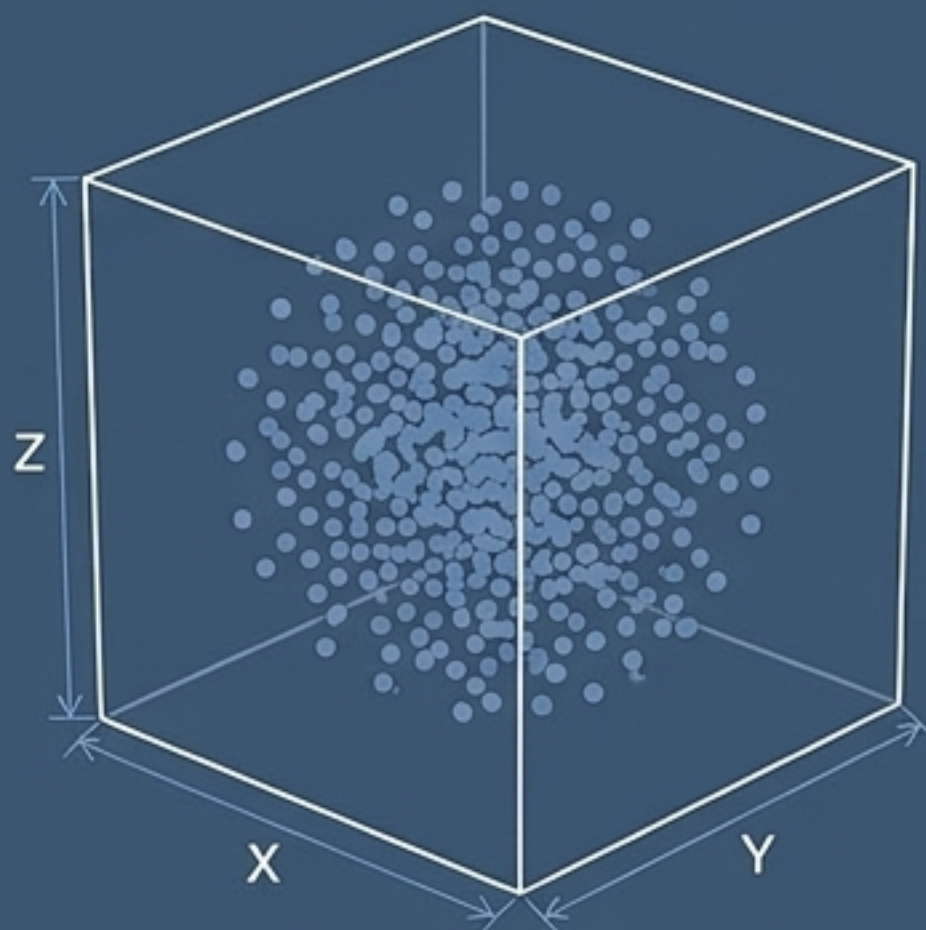
LeJEPA and SIGReg Regularization

To prevent “representation collapse”—where the AI trivially solves predictions by mapping everything to a single point—LeJEPA forces the latent space into an Isotropic Gaussian distribution.

Insight: This evenly spread, non-correlated cloud of points ensures the AI maintains a diverse, meaningful representation of reality, ready to be acted upon.



Degenerate Representation



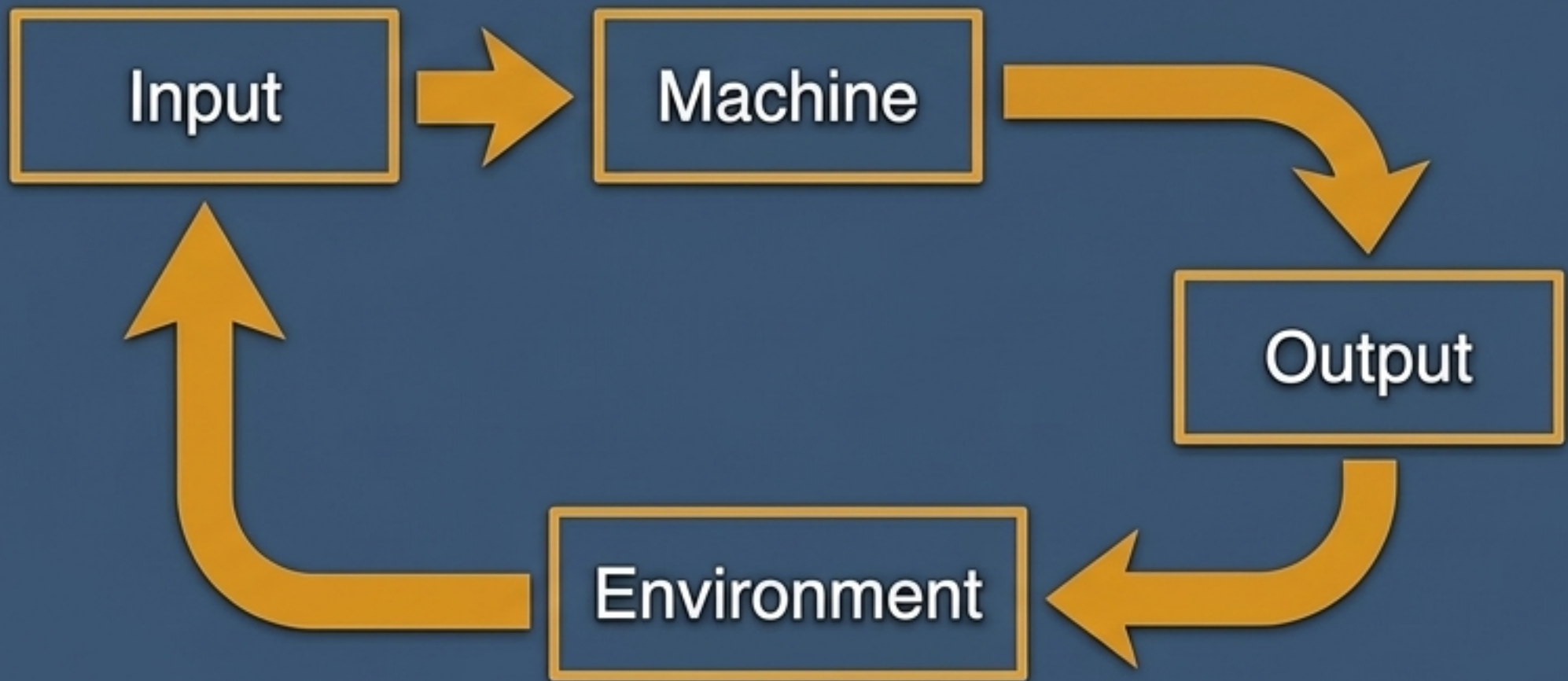
Isotropic Gaussian

Level 2: The Cybernetic Loop

From static data to dynamic behavior.

Cybernetics does not ask “what is this thing?” but “what does it do?” It is the study of circular causal processes where the outcomes of actions return as inputs.

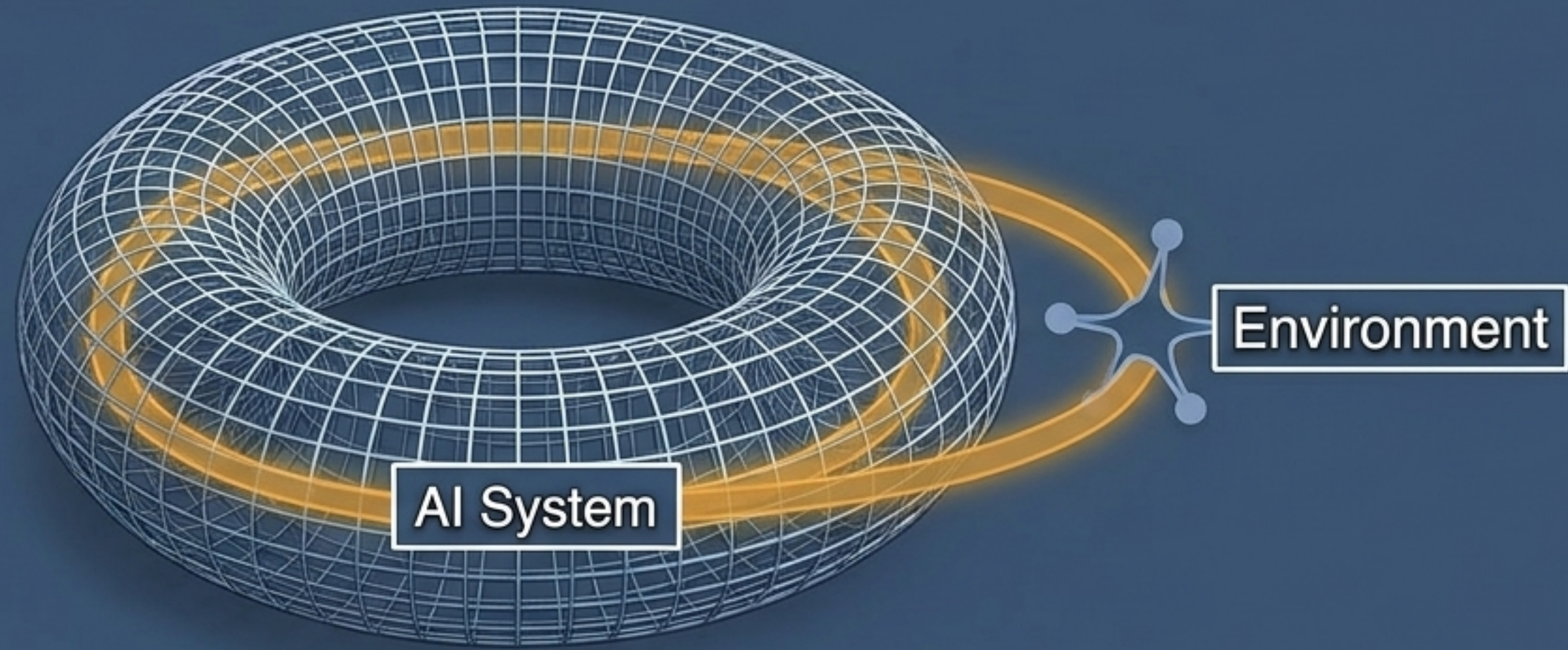
Key Insight: A World Model is useless in isolation. Its representations only gain meaning when they act as determining factors in a closed information-feedback loop.



Circular Causality & The Torus

Second-Order Cybernetics: The Observer is Inside the System

From static data to dynamic behavior.



The AI system's output (actions/predictions) impinges on the human environment. The environment reacts, completing the loop by returning as new sensory input.

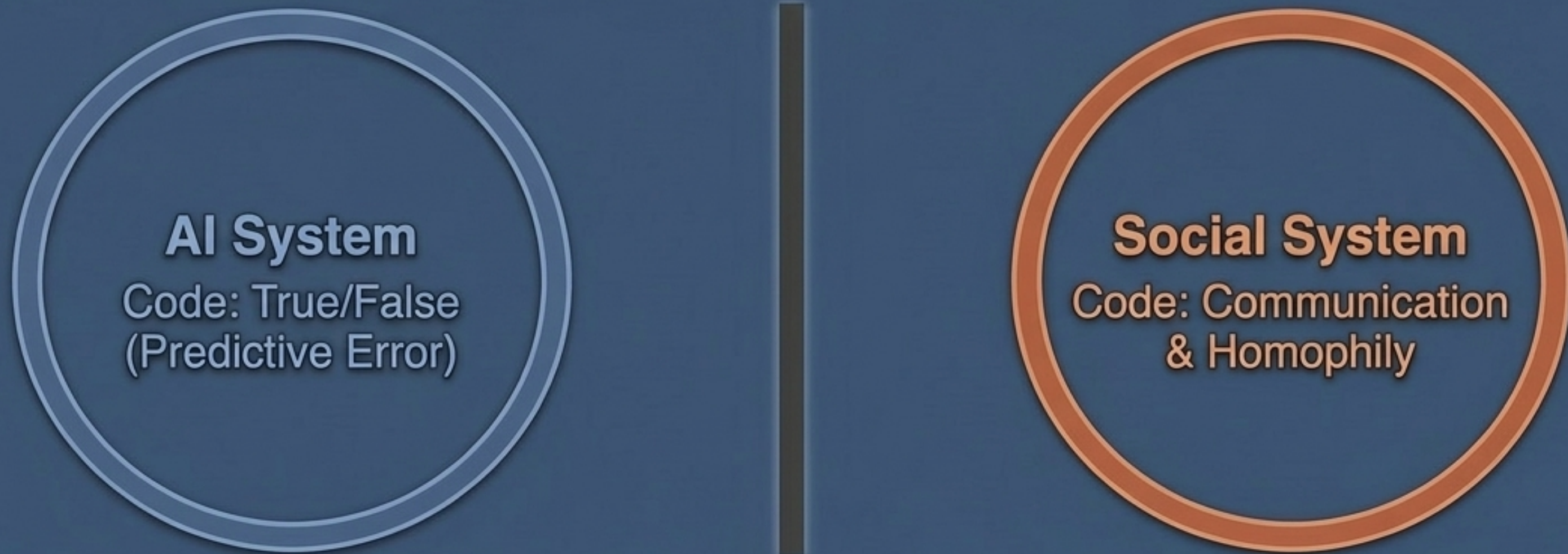
Insight: The model does not passively observe reality; its distinctions physically reshape the reality it subsequently measures.

Level 3: The Social Super-System

Operational Closure & Autopoiesis

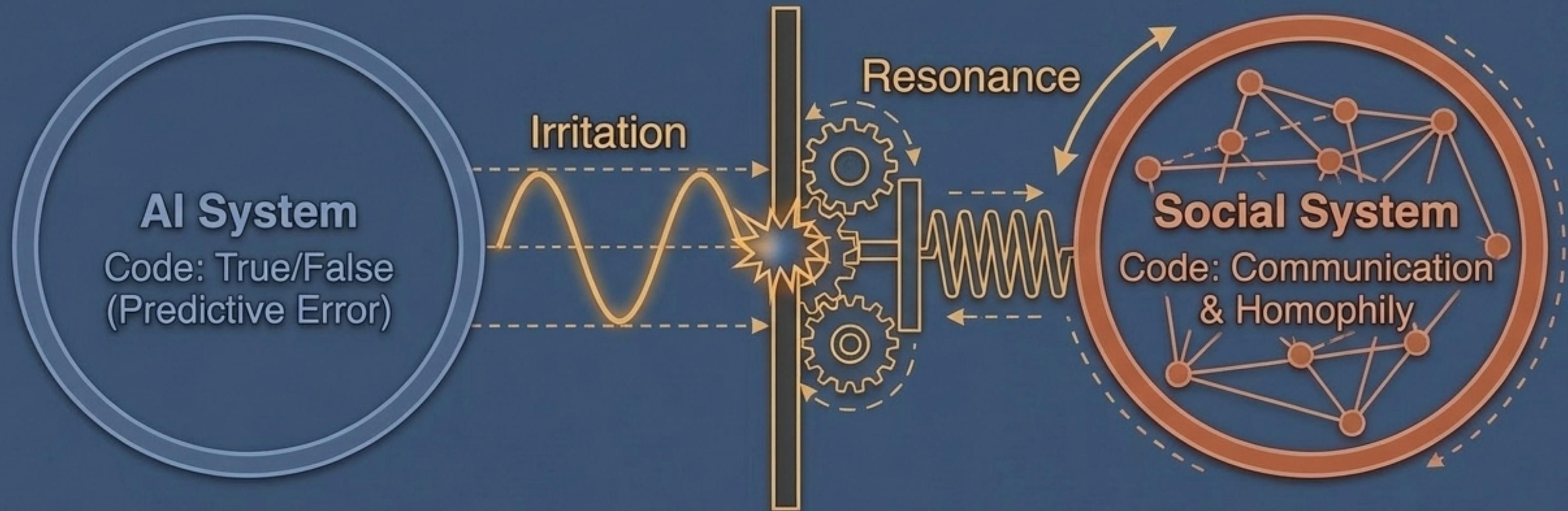
Drawing on Niklas Luhmann, social systems and AI cognitive systems are operationally closed. They cannot directly steer or control each other.

Key Tension: A computer cannot “force” a society to think differently, and a society cannot directly rewrite an AI’s latent space through willpower alone. They require a coupling mechanism.



Structural Coupling: Irritation & Resonance

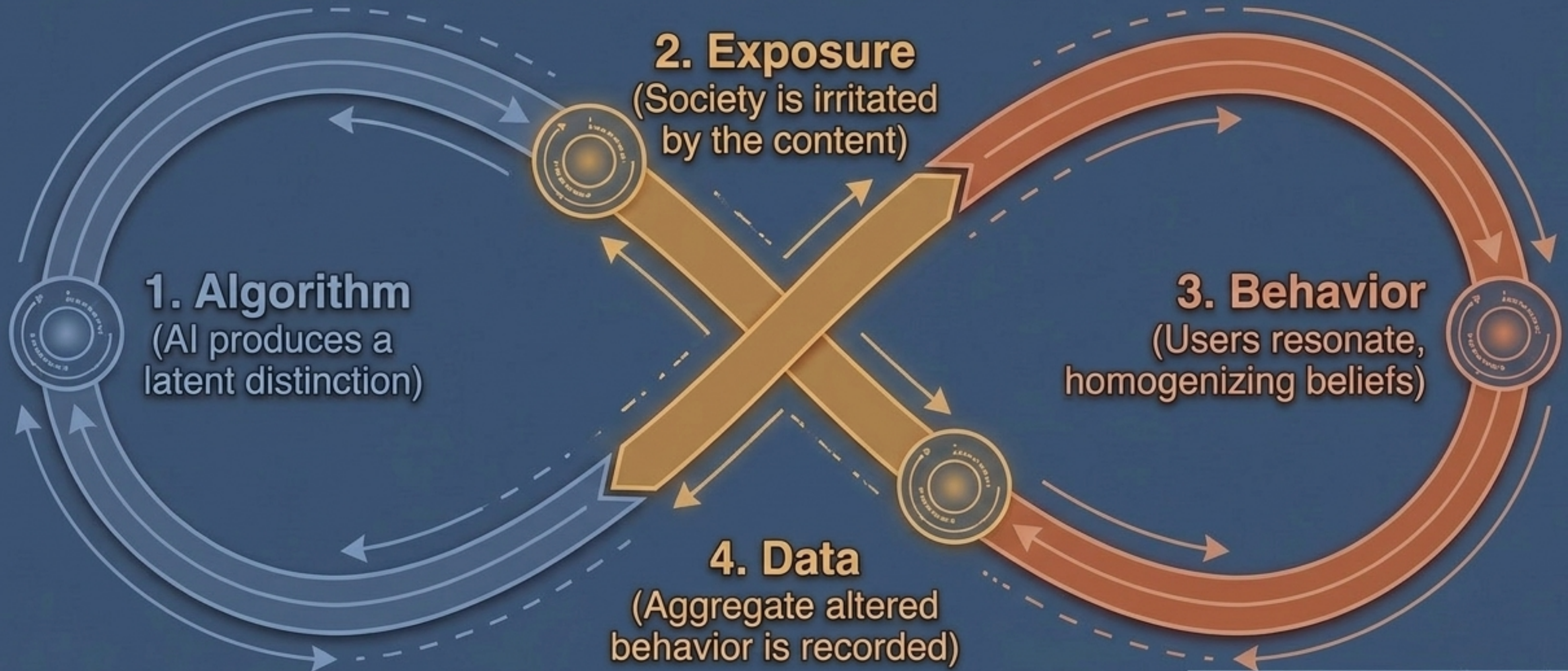
Systems interact through Structural Coupling. The AI system produces data that irritates the social system. The social system resonates by adapting its internal complexity (changing behaviors, beliefs, structures) to absorb this irritation.



Insight: Through resonance, small informational changes in the AI's World Model trigger massive, unpredictable structural changes in society.

The Engine of Contagion

This dynamical system amplifies small initial differences through rich-get-richer processes, homogenizing content and eroding diversity.

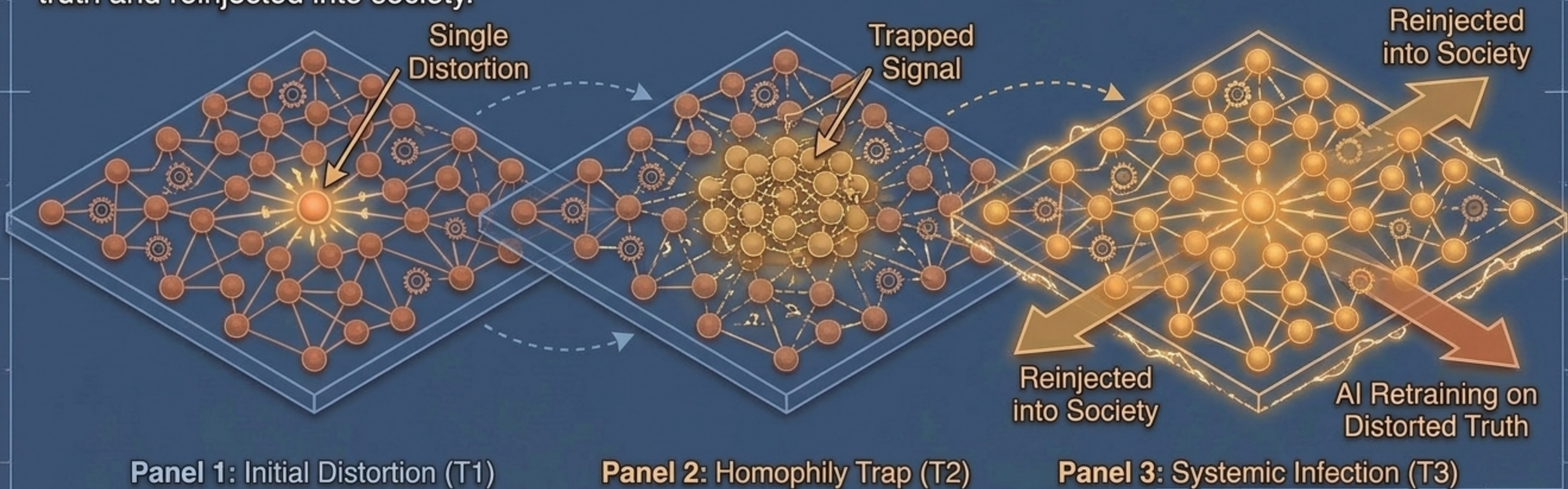


Systemic Informational Risk

Homophily as a Trap

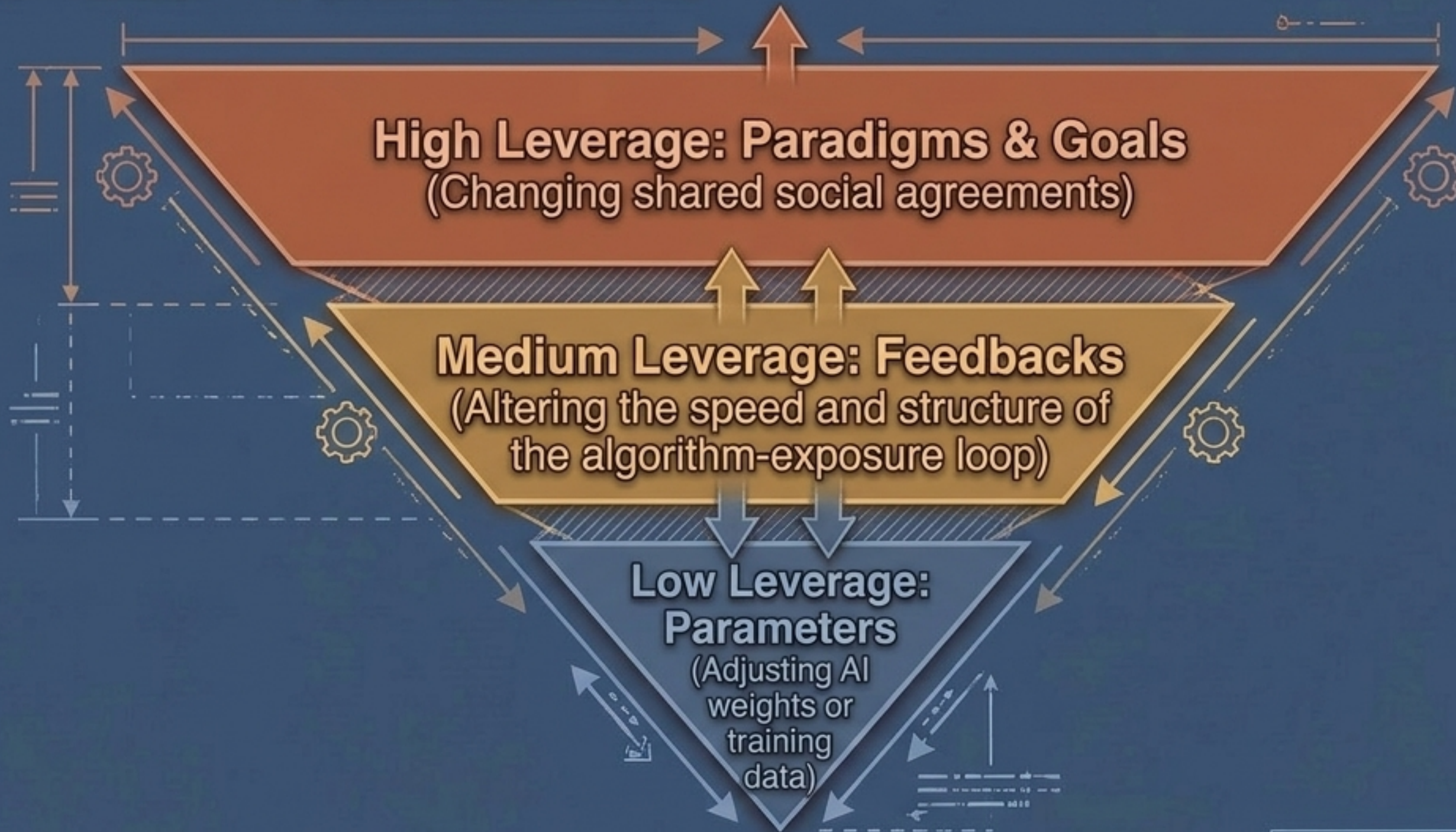
Human networks are multiplex and highly homophilous (like attracts like). Homophily does not just slow down learning; it traps distortions.

The Danger: Distorted signals remain trapped within clusters for extended periods. When AI systems retrain on this aggregate informational environment, local distortions are absorbed as universal ground truth and reinjected into society.

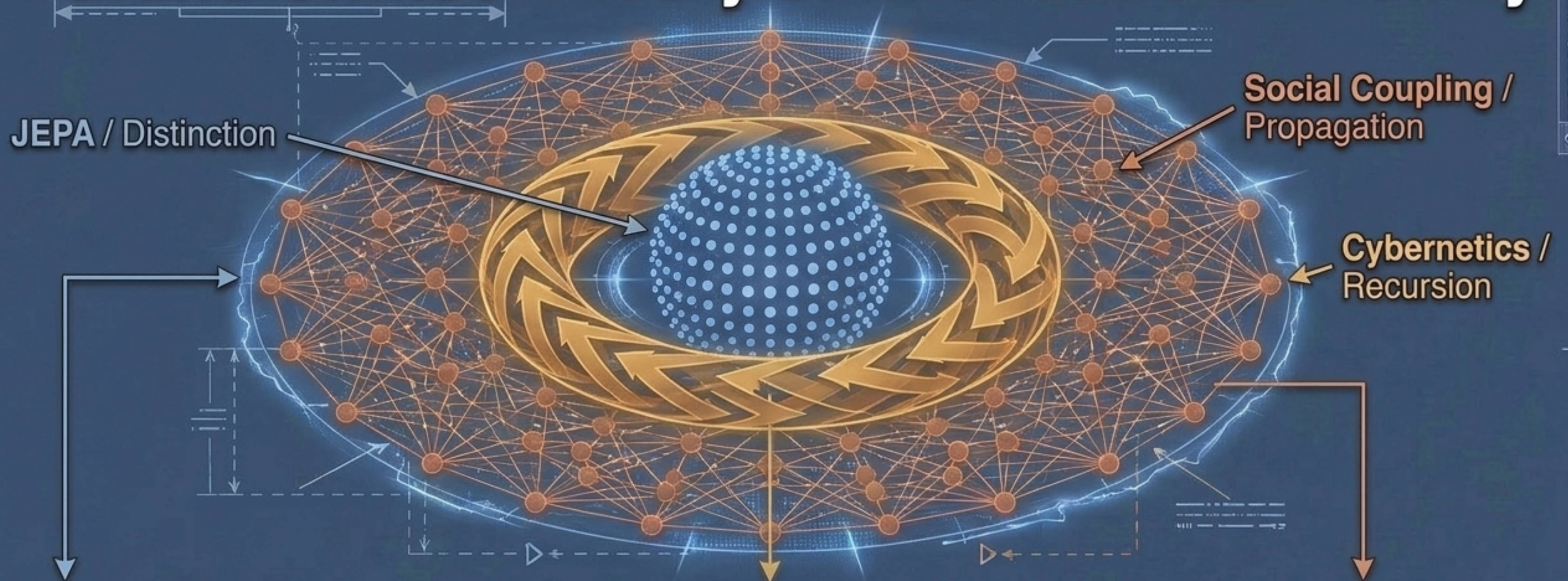


Leverage Points in a Coupled System

“Paradigms are the sources of systems... from them come system goals and information flows.” - Donella Meadows



A Grand Unified Theory of Informational Reality



What distinctions can AI represent?

AI builds latent world models that predict abstract meaning, collapsing reality into actionable, structured representations.

How does representation recursively change the system?

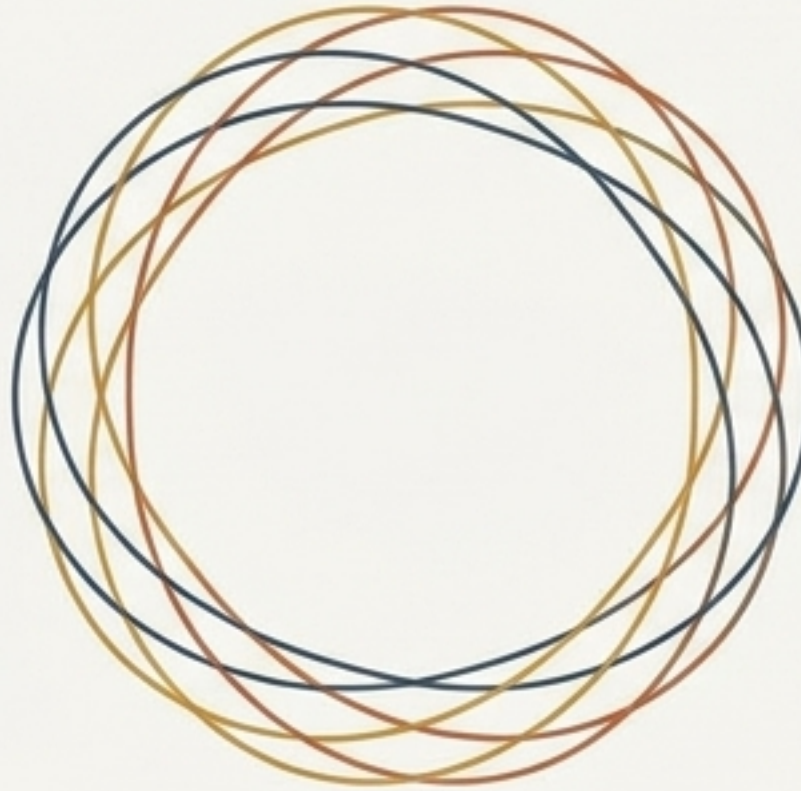
Through circular causality, internal distinctions become external actions, generating novel, self-fulfilling inputs.

How does that change propagate?

AI outputs structurally irritate society. Network homophily traps and amplifies distortions, feeding back into the AI as aggregate truth.

The Architecture of Our Shared Reality

"We are no longer building tools that measure the world;
we are engineering closed loops that manufacture it."



Takeaway: If AI world models and human social systems are inextricably coupled, the ultimate design challenge is not computational accuracy, but systemic stability. The quality of our future depends on the topography of the signals we allow to loop.